Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Digital Assignment Lab - 3

Course Name :- Agile Development Process and Devops Lab

Course code :- ISWE406P

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Reg-No:- 23MIS0227

1- Create a Maven Project in Git

• Create a new Git repository

```
PS C:\Users\saisa\OneDrive\Desktop\MavenGitProject> git init
Initialized empty Git repository in C:/Users/saisa/OneDrive/Desktop/MavenGitProject/.git/
PS C:\Users\saisa\OneDrive\Desktop\MavenGitProject> mkdir -p src/main/java/com/example


    Directory: C:\Users\saisa\OneDrive\Desktop\MavenGitProject\src\main\java\com


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        02-03-2026     20:35                example
```

• Create a simple Maven project structure manually, including the following files and folders:

jenkinsfile

pom.xml

src/main/java/com/example/App.java

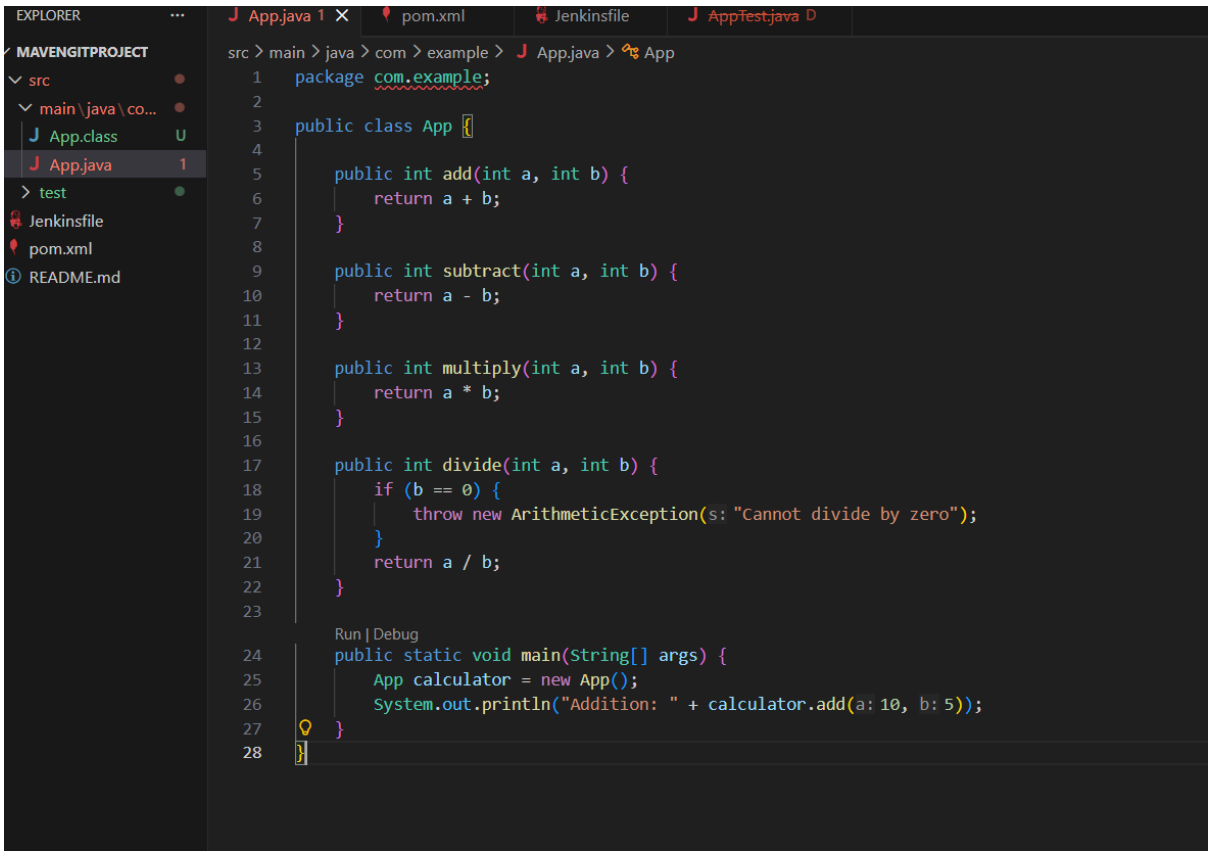
src/test/java/com/example/AppTest.java

```
PS C:\Users\saisa\OneDrive\Desktop\MavenGitProject> tree /F
Folder PATH listing for volume OS
Volume serial number is 686E-F4A0
C:.
│   Jenkinsfile
│   pom.xml
│   README.md
│
└───src
    ├───main
    │   └───java
    │       └───com
    │           └───example
    │                   App.java
    │
    └───test
        └───java
            └───com
                └───example
                        AppTest.java
```

2- Implement the Java Program

• Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios:

Simple Calculator Application

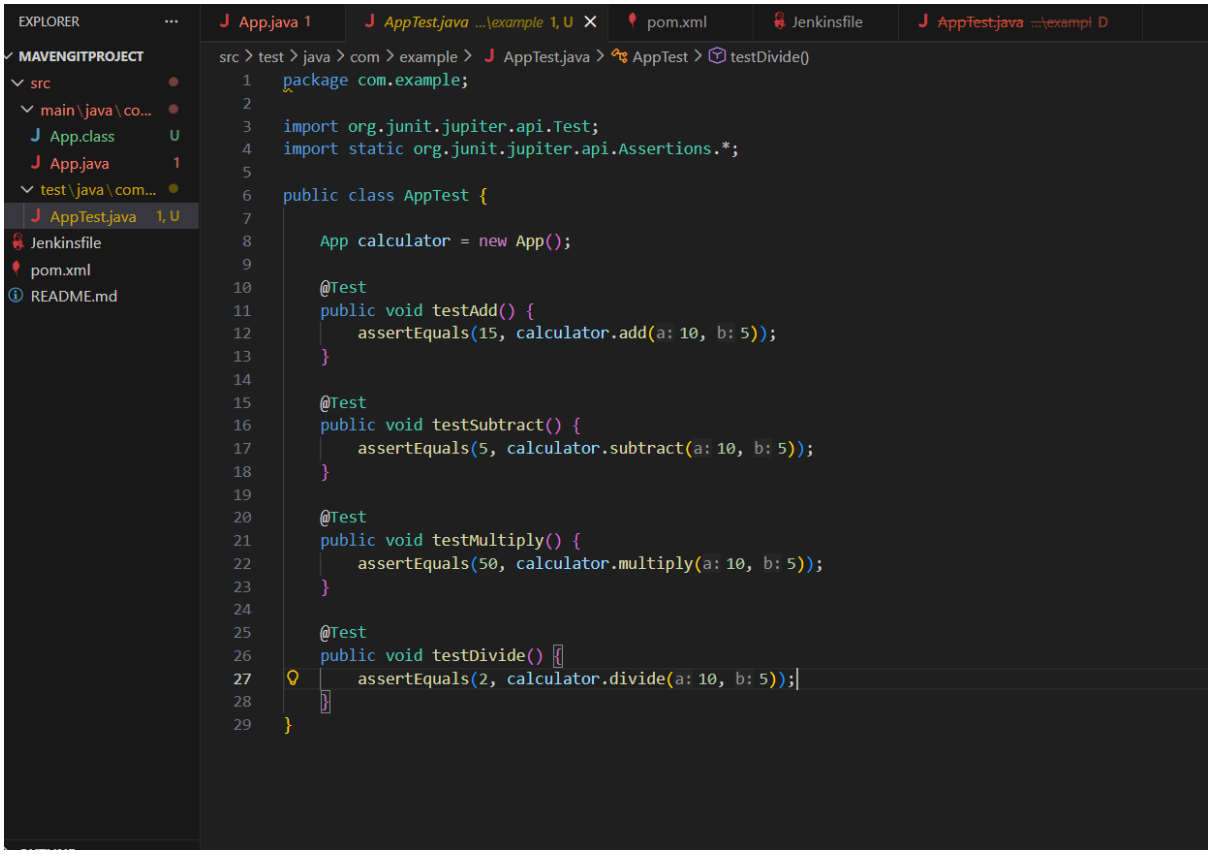


Code :

```
package com.example;

public class App {

    public int add(int a, int b) {
        return a + b;
```

```
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public int divide(int a, int b) {
        if (b == 0) {
            throw new ArithmeticException("Cannot divide by zero");
        }
        return a / b;
    }

    public static void main(String[] args) {
        App calculator = new App();
        System.out.println("Addition: " + calculator.add(10, 5));
    }
}
```

- Write a corresponding unit test in AppTest.java.



Code :

```
package com.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class AppTest {

    App calculator = new App();
```

```
    @Test
    public void testAdd() {
        assertEquals(15, calculator.add(10, 5));
    }

    @Test
    public void testSubtract() {
        assertEquals(5, calculator.subtract(10, 5));
    }

    @Test
    public void testMultiply() {
        assertEquals(50, calculator.multiply(10, 5));
    }

    @Test
    public void testDivide() {
        assertEquals(2, calculator.divide(10, 5));
    }
}
```

Code for Pom.xml:

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>calculator-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>
        <!-- JUnit 5 -->
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.9.3</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.0.0-M5</version>
            </plugin>
        </plugins>
    </build>

</project>
```

3- Push and Clone the Repository

- Clone the repository using Visual Studio Code.

```
PS C:\Users\saisa\OneDrive\Desktop\MavenGitProject> git add .
>> git commit -m "Initial Maven Calculator Project"
[master (root-commit) e43a297] Initial Maven Calculator Project
 3 files changed, 89 insertions(+)
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/exampl/AppTest.java
```

```
PS C:\Users\saisa\OneDrive\Desktop\MavenGitProject> git pull origin main --allow-unrelated-histories
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 867 bytes | 173.00 KiB/s, done.
From https://github.com/santhoshh005/MavenGitProject
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
PS C:\Users\saisa\OneDrive\Desktop\MavenGitProject> git push -u origin main
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (16/16), 1.82 KiB | 932.00 KiB/s, done.
Total 16 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/santhoshh005/MavenGitProject.git
   d83758a..2737aec  main -> main
branch 'main' set up to track 'origin/main'.
```

- Verify the project structure.

```
PS C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject> tree /f
Folder PATH listing for volume OS
Volume serial number is 686E-F4A0
C:.
│   pom.xml
│   README.md
│
├───src
│   ├───main
│   │   └───java
│   │       └───com
│   │           └───example
│   │               App.java
│   │
│   └───test
│       └───java
│           └───com
│               └───example
│                   AppTest.java
│
└───target
    │   calculator-app-1.0-SNAPSHOT.jar
    │
    ├───classes
    │   └───com
    │       └───example
    │           App.class
```

```
│   ├──generated-sources
│   │   └──annotations
│   ├──generated-test-sources
│   │   └──test-annotations
│   ├──maven-archiver
│   │       pom.properties
│   │
│   ├──maven-status
│   │   └──maven-compiler-plugin
│   │       ├──compile
│   │       │   └──default-compile
│   │       │           createdFiles.lst
│   │       │           inputFiles.lst
│   │       │
│   │       └──testCompile
│   │           └──default-testCompile
│   │               createdFiles.lst
│   │               inputFiles.lst
│   │
│   ├──surefire-reports
│   │       com.example.AppTest.txt
│   │       TEST-com.example.AppTest.xml
│   │
│   └──test-classes
│       └──com
│           └──example
│               AppTest.class
```

4- Build the Maven Project

- Use Maven commands in Visual Studio Code terminal to:

Compile the project

Command Used:- mvn compile

```
PS C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:calculator-app >--------------------
[INFO] Building calculator-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ calculator-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.400 s
[INFO] Finished at: 2026-03-02T21:03:04+05:30
[INFO] ------------------------------------------------------------------------
```

Run tests

Command Used:- mvn test

```
PS C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:calculator-app >--------------------
[INFO] Building calculator-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ calculator-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ calculator-app ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.0.0-M5:test (default-test) @ calculator-app ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
```

```
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.03 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.557 s
[INFO] Finished at: 2026-03-02T21:03:10+05:30
[INFO] ------------------------------------------------------------------------
```

Build the project (mvn clean install)

Command Used:- mvn clean install

```
PS C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject> mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:calculator-app >--------------------
[INFO] Building calculator-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.
4.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.4
.1.pom (7.8 kB at 14 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/42/maven-plugins-42.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/42/maven-plugins-42.pom (7.7
kB at 96 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/42/maven-parent-42.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/42/maven-parent-42.pom (50 kB at 934 k
B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/32/apache-32.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/32/apache-32.pom (24 kB at 145 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.2/maven-install-p
lugin-3.1.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.2/maven-install-pl
ugin-3.1.2.pom (8.5 kB at 75 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ calculator-app ---
[INFO] Deleting C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ calculator-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
```

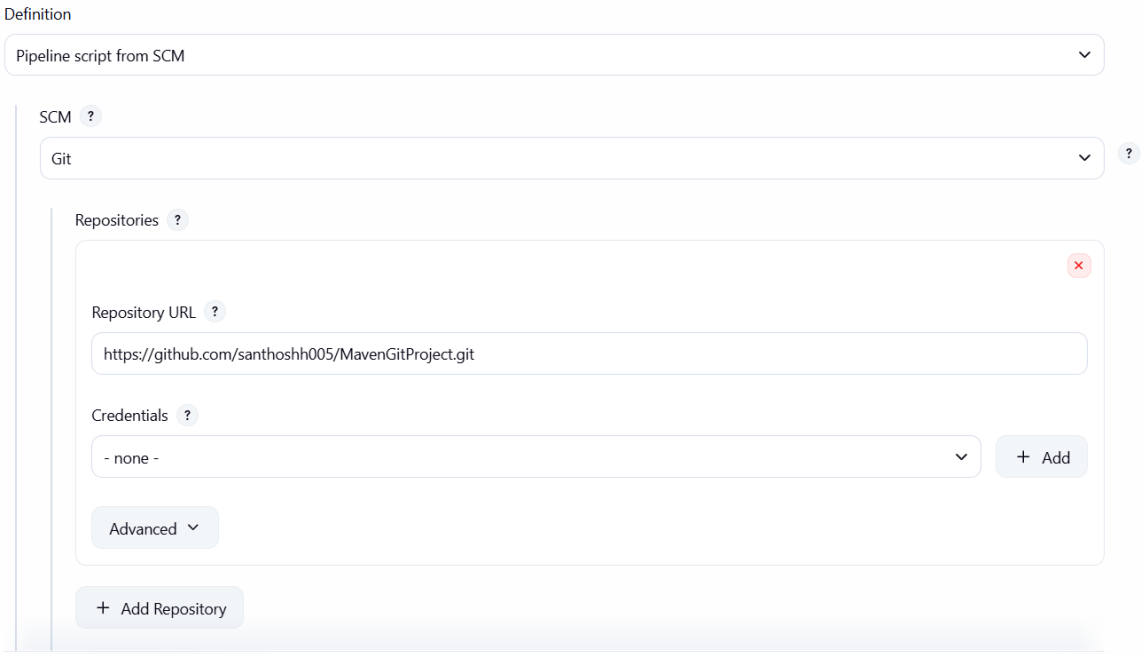
```
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ calculator-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.0.0-M5:test (default-test) @ calculator-app ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.029 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ calculator-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.1.0/file-management-3.1.0
.pom
```

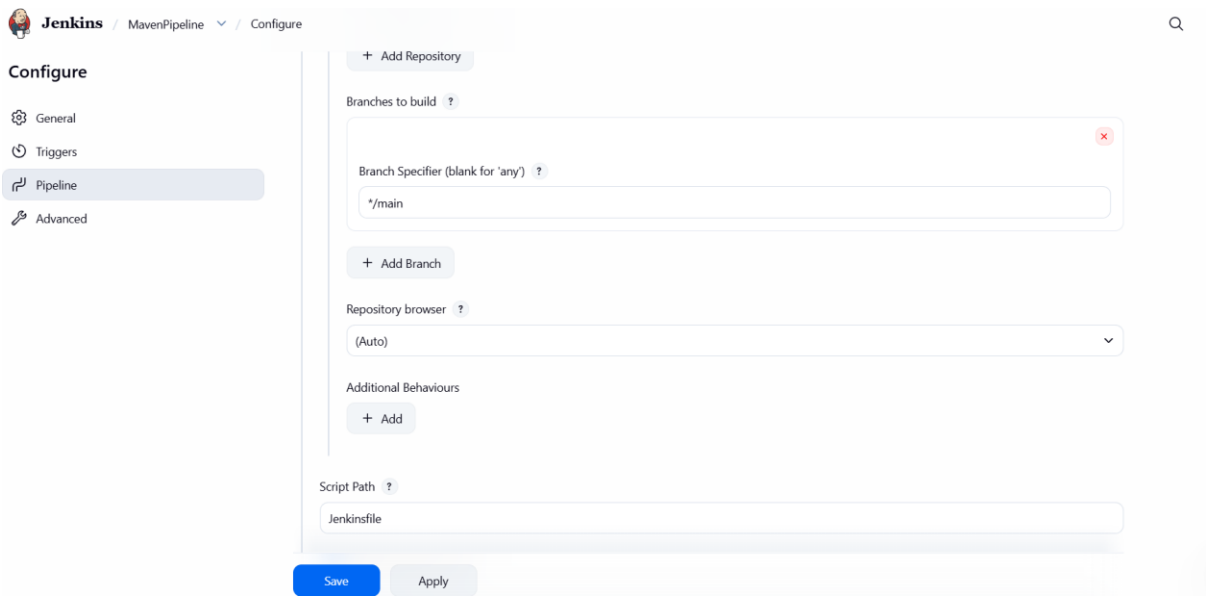
5- Create and Run Jenkins Pipeline

- Create a new Pipeline project in Jenkins.
- Create a pipeline script

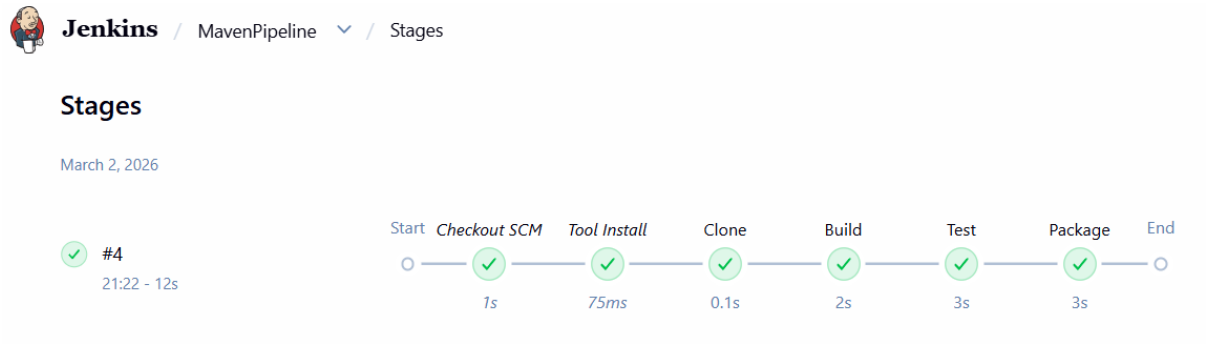
```
pipeline {
    agent any

    stages {

        stage('Clone Repository') {
            steps {
```

```
            git 'https://github.com/yourusername/MavenGitProject.git'
        }
    }

    stage('Build') {
        steps {
            bat 'mvn clean install'
        }
    }
  }
}
```





- Execute the pipeline and verify successful build completion.



Jenkins / MavenPipeline

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Stages
- Rename
- Pipeline Syntax

Builds

Filter /

Today

#4 9:22 PM

MavenPipeline

Permalinks

- Last build (#4), 38 min ago
- Last stable build (#4), 38 min ago
- Last successful build (#4), 38 min ago
- Last completed build (#4), 38 min ago